

Archaeologist

14+3 · SIBLING placement vs Meta chain

Three branches: Meta SSL I-JEPA→...→2-AC · Control E2E PLDM→LeWM · Foundation freeze DINO-WM. Do not contradict prior CLEAR labels — extend.

SPINE: Stable E2E JEPA WM · MSE+SIGReg only · ~15M · 48× vs DINO-WM (0.98s/47s) · SIBLING to Meta JEPA · SIM-ONLY · PushT 96 ± 2.83 · fixed-FLOP 90vs13 · recon hurts · λ -only

Talk arc — Archaeologist

Priors → inventum → three branches → labels

02-04

Spine + CLEAR priors

AMI · I/V-JEPA · PLDM · DINO-WM · VICReg · LeJEPA

05-07

Inventum + three branches

two-term E2E · Meta vs Control vs Freeze

08-10

Triangle + subsequent

CLEAR code · TENTATIVE Franka/BEHAVIOR/ $\pi_{0.5}$

11-14

Labels + tattoo + close

CLEAR vs TENTATIVE exam discipline

Shared spine — tattoo every deck

Five claims this paper adds to the club

1. Stable E2E JEPA WM from pixels — MSE + SIGReg only; drop EMA/SG/freeze/multi-VICReg.
2. ~15M params; plan ~48× faster than DINO-WM (0.98s vs 47s).
3. SIBLING to Meta JEPA line (not descendant). SIM-ONLY — cite V-JEPA 2-AC for real-robot JEPA planning.
4. Must: PushT 96 ± 2.83 ; fixed-FLOP 90 vs 13; recon hurts (96→86); TwoRoom behind DINO; λ -only knob.
5. Foils: PLDM (fragile multi-term) · DINO-WM (frozen foundation) · Meta 2-AC (real robot different recipe).

CLEAR priors

Cited / direct parents — do not demote

LeCun 2022 AMI	Names JEPA agenda	CLEAR
I-JEPA (2301.08243)	Image JEPA; emb pred; EMA/SG	CLEAR
V-JEPA / V-JEPA 2	Video JEPA; scale; planning mention	CLEAR
PLDM (Sobal 2025)	E2E JEPA-WM · VICReg multi-term	CLEAR foil
DINO-WM (Zhou 2025)	Frozen DINOv2 WM + CEM	CLEAR foil
VICReg (Bardes)	PLDM regularization ancestor	CLEAR
LeJEPA / SIGReg (2511.08544)	Provable Gaussian anti-collapse	CLEAR substrate
Dreamer / TD-MPC / IRIS	Generative / reward / pixel foils	CLEAR related

CLEAR inventum — what THIS paper adds

Sibling recipe, not Meta child

Two-term E2E

First claimed stable end-to-end JEPA from raw pixels with MSE + SIGReg only.

λ -only tuning

Removes EMA, stop-grad, frozen encoder, and PLDM's six+ loss weights.

15M + 48×

Compact AC world model with $\sim 48\times$ planning speedup vs DINO-WM + strong Push-T.

Physics suite

Probing + VoE + emergent temporal straightening without temporal loss.

Open code

[lucas-maes/le-wm](https://lucas-maes.github.io) + le-wm.github.io — CLEAR reproducibility path.

Three branches — SIBLING placement

CLEAR labels on every column

Meta SSL JEPA

CLEAR

I-JEPA → V-JEPA →
V-JEPA 2 → 2.1 → 2-AC

Franka CEM
(CLEAR robot path)

EMA/SG + scale

Control E2E JEPA-WM

CLEAR

PLDM (fragile)
→ LeWM (THIS)

Sim continuous
control CEM

MSE+SIGReg
 λ -only

Foundation freeze WM

CLEAR

DINO-WM

Frozen DINOv2
+ predictor

No collapse via freeze
~200× tokens

Fig. 2 — taxonomy evidence

Paper's own grouping of latent WM paradigms



Figure 2: **Characteristics of latent world model approaches.** Methods are grouped by training paradigm. *End-to-end* methods (PLDM) learn the encoder and predictor jointly from pixels without pre-trained representations or heuristics like stop-gradient or EMAs, but require many hyperparameters and lack collapse guarantees. *Foundation-based* methods (DINO-WM) avoid collapse by freezing a pre-trained vision encoder, forgoing end-to-end learning. *Task-specific* methods (Dreamer, TD-MPC) require reward signals or privileged state access. LeWM combines the strengths of all three: end-to-end, task-agnostic, pixel-based, reconstruction- and reward-free, with a single hyperparameter with provable anti-collapse guarantees.

1 Introduction

A central goal of artificial intelligence is to develop agents that acquire skills across diverse tasks and environments using a single, unified learning paradigm—one that operates directly from sensory inputs of its surroundings—without hand-engineered state representations or domain-specific calibration. Vision is ideally suited for this aim: cameras are inexpensive and scalable, and learning from pixels enables fully end-to-end training from raw sensory input to action [1]. World Models (WMs) are a powerful family of methods [2] that learn to predict the consequences of actions in the environment. When successful, WMs allows agents to plan and to improve themselves solely from their model of the world, i.e., in imagination space. This is particularly valuable in the offline setting, where agents must learn from fixed datasets without environment interaction—leveraging the model to generate synthetic experience and evaluate counterfactual action sequences [3, 4].

A recent popular approach for learning world models is the Joint Embedding Predictive Architecture (JEPA) [5]. Instead of attempting to model every aspect of the environment, JEPA focuses on

Use as visual for three-branch placement; LeWM in E2E JEPA column.

Consistency triangle — extended

Do not contradict I-JEPA / V-JEPA 2.1 packs

DINOv2 / SigLIP

VLA towers / DINO-WM freeze

discriminative JEA / dense features

MAE / Dreamer / IRIS

FT or RL imagination

generative pixels / reward-tied

I-JEPA → ... → 2-AC

real-robot CEM (CLEAR)

Meta predictive SSL → AC plan

PLDM → LeWM

fast sim planning (THIS)

compact E2E JEPA-WM + SIGReg

Subsequent — CLEAR vs TENTATIVE

Label discipline on lineage slides

CLEAR subsequent

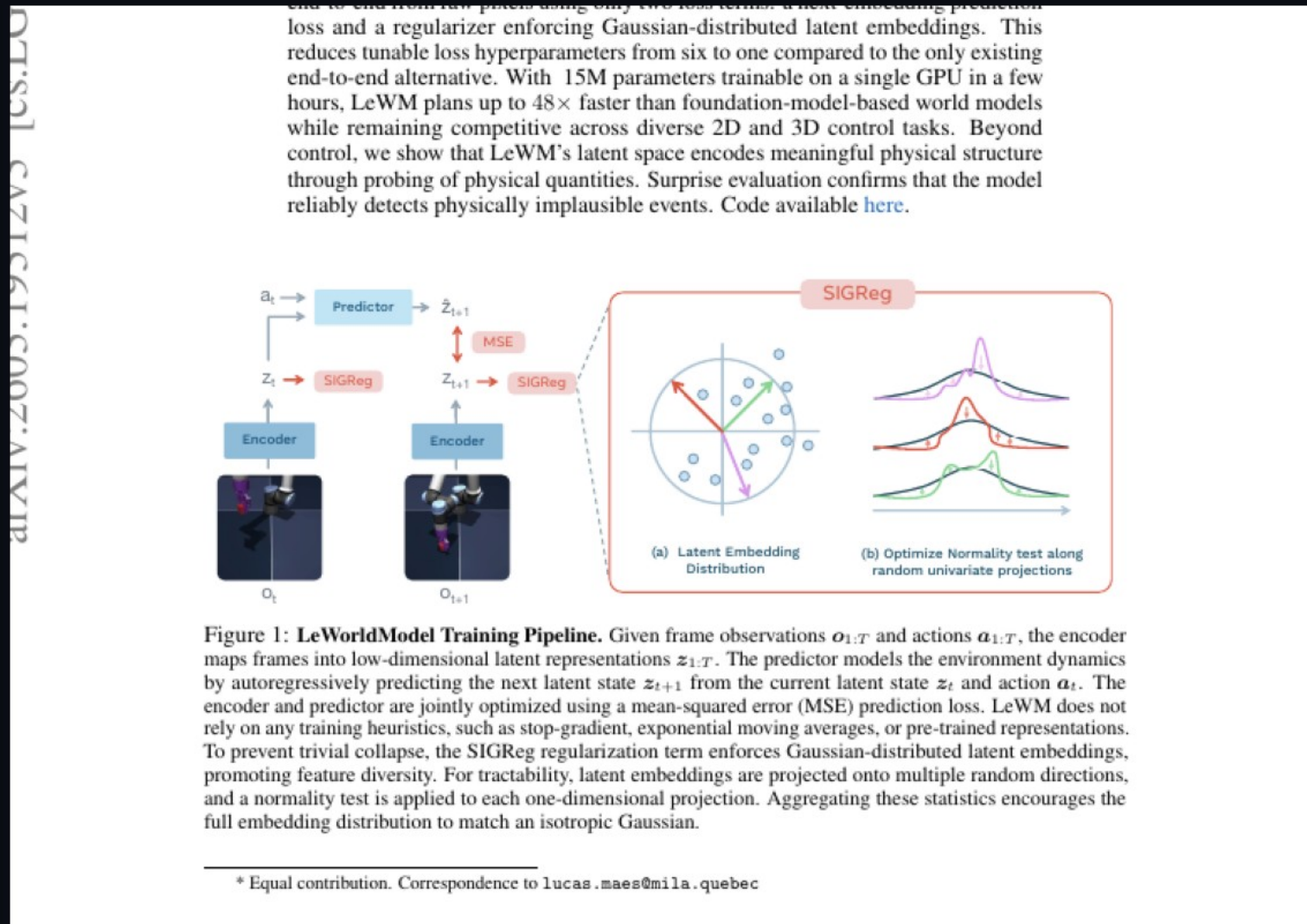
- lucas-maes/le-wm + le-wm.github.io
- Prior-pack Meta chain I-JEPA→...→2-AC remains CLEAR (do not demote)
- V-JEPA 2.1 dense unlock CLEAR in prior pack (not cited here — parallel)

TENTATIVE / parallel

- Real Franka / BEHAVIOR with LeWM weights — not shown
- Merge SIGReg E2E with V-JEPA 2.1 dense features — open
- $\pi_{0.5}$ + FM stacks — parallel generative-action bet (rhyme only)
- Causal-JEPA / OSVI-WM — parallel JEPA-WM threads

Fig. 1 — inventum visual

CLEAR: MSE + SIGReg E2E pipeline from pixels



Architecture inventum slide — no EMA target drawn.

CLEAR vs TENTATIVE — exam discipline

Say the label before the claim

CLEAR

LeWM is sibling control-WM recipe to Meta SSL line

CLEAR

I-JEPA→V-JEPA→2→2.1→2-AC remains the Meta robotics path

CLEAR

PLDM & DINO-WM are the primary foils

TENTATIVE

LeWM on real Franka / BEHAVIOR

TENTATIVE

SIGReg predictor on V-JEPA 2.1 features

TENTATIVE

$\pi_{0.5}$ hybrid rhyme (not cited by LeWM)

Foils mnemonic

Three names to keep straight

PLDM

Fragile multi-term VICReg E2E

Closest sibling · LeWM
replaces loss cocktail

DINO-WM

Frozen foundation encoder

No collapse via freeze ·
slow tokens

V-JEPA 2-AC

Real-robot different recipe

Cite for Franka — not a LeWM
descendant

Numbers tattoo — pin these

Shared spine across all five seats

~15M

params · 1 GPU / few h

0.98s vs 47s

~48× planning speed

~200×

fewer tokens vs DINO

90 vs 13

fixed-FLOP Push-T

74 vs 48

fixed-FLOP Cube

96.0±2.83

Push-T SR (Tab.5)

78±5 / 92±1.63

PLDM / DINO Push-T

$\lambda \in [0.01, 0.2]$

SR>80%; peak ~0.09

96 → 86

recon loss hurts (T7)

SIBLING

not Meta descendant

SIM-ONLY

cite 2-AC for robots

λ -only

vs PLDM 6 knobs

Archaeologist — close

Takeaways + exam bite

1. Placement: SIBLING control E2E JEPA-WM — not Meta descendant.
2. CLEAR priors: AMI, I/V-JEPA, PLDM, DINO-WM, VICReg, LeJEPA/SIGReg.
3. Inventum: two-term E2E · λ -only · 15M+48× · physics suite · open code.
4. TENTATIVE: real Franka/BEHAVIOR; merge with 2.1; $\pi_{0.5}$ rhyme.

EXAM BITE

LeWM = alternative JEPA training recipe for compact AC WMs from pixels. Sibling to PLDM; foil to DINO-WM and to Meta freeze-then-AC. Not a replacement for V-JEPA 2-AC's real-robot result.